JOC Computational Project: Milestone 1 Report

Phase Objective: 1/5 — Structural Library Curation & Conformer Coordinate Generation
Status: **100% COMPLETED**

1. Methodological Justification & Scope

We have curated a highly diversified library of terminal alkynes to track molecular ground-state polarization. The components represent key physical-organic environments: pure inductive vectors (+I and -I via alkyl and halogenated fragments) and active mesomeric networks (+R and -R via para-substituted benzene derivatives). This robust, balanced dataset fulfills the strict structural breadth guidelines expected by the reviewers of *The Journal of Organic Chemistry*, preventing statistical overfitting or initial collection bias.

2. Computational Workflow Executed

Each terminal alkyne structure was modeled atom-by-atom in a three-dimensional coordinate matrix. Conformational space minimization was achieved using the MMFF94 molecular mechanics force field to calculate structural ground-state geometry configurations. The structural geometries have been exported directly into clean, standardized Cartesian coordinate text frames (.xyz). These files will serve as input matrices for Phase 2.

3. Curated Library Index Data Table

Compound ID	Substituent (-R)	XYZ Filename	Electronic Archetype
1a	Methyl	1a_propyne.xyz	+I (Weak Inductive)
1b	tert-Butyl	1b_tert_butylacetylene.xyz	+I (Strong Inductive/Steric)
1c	Trifluoromethyl	1c_trifluoromethylacetylene.xyz	-I (Strong Inductive Withdrawing)
1d	Phenyl	1d_phenylacetylene.xyz	Conjugated (Aryl Baseline)
1e	p-Nitrophenyl	1e_p_nitrophenylacetylene.xyz	-R / -I (Resonance Withdrawing)
1f	p-Methoxyphenyl	1f_p_methoxyphenylacetylene.xyz	+R (Resonance Donating)
1g	Cyano	1g_cyanoacetylene.xyz	-R / -I (Linear Core Withdrawing)